LBSM Research — Notebook 07

Robustness & Final Experiment Analysis

Full-Scale `LBSM-ISSUE-NB07-001` Protocol: 160-Configuration Robustness Grid, Extended BIC Model-Order Search, and HMM State-Identity Diagnostics

Grid	$N \in \{5, 10, 20, 50\}$, $T \in \{500, 1{,}000, 2{,}000, 5{,}000\}$, 10 seeds (42–51) — 160 configs, the complete un-reduced `LBSM-ISSUE-NB07-001` protocol
Per-config fitting	diagonal baseline (`fit_hmm`) + full covariance (`fit_hmm_robust`)
Full-covariance attempts/config	3 `min_covar` values $\times$ 10 restarts = 30 attempts (4,800 full-covariance fits across the whole grid)
BIC model-order search	$K = 2, \ldots, 8$ (widened from the earlier $K = 2, \ldots, 6$)
State-identity diagnostic	N=50, T=5,000, seed=42 (250,000 observations, the largest configuration in the grid)
Execution date	2026-08-26
Environment	Python 3.13.12 (Nix flake dev shell) / Jupyter (ipykernel)
Key package versions	numpy 2.4.4, scipy 1.17.1, scikit-learn 1.8.0, hmmlearn 0.3.3
Random seeds	42–51 (grid); 42 (state-identity diagnostic)

Central Question: NB03's diagonal-covariance HMM could not recover the correct model order (BIC selected $K{=}6$, not the ground-truth $K{=}4$) because the three healthy regimes form a continuous curved manifold (NB02) that an axis-aligned Gaussian cannot represent. Does switching to full covariance — which can represent that correlation structure — recover the correct BIC elbow and improve regime-recovery accuracy, and does it do so robustly across a realistic range of dataset sizes, or does numerical instability at small $N \times T$ undermine the results, as anticipated in `LBSM-ISSUE-NB07-001`? And if full covariance does not fix the BIC over-selection either, *why not* — is there a specific, testable, mechanistic explanation, and does the answer change once the model-order search is not artificially bounded?

This document records the full methodology, numerical results, figures, and mechanistic investigation produced by the seventh (final analytical) notebook of the LBSM research project, **executed at full scale**: the complete, un-reduced `LBSM-ISSUE-NB07-001` protocol (160 configurations, 4,800 full-covariance fits, every individual attempt logged), superseding an earlier compute-budget-reduced pass (24 configurations, $T \leq 2{,}000$, $K \leq 6$) that was used to validate the mitigation stack and infrastructure before committing to full scale. Widening the model-order search from $K = 2, \ldots, 6$ to $K = 2, \ldots, 8$ changed the central finding materially: BIC's over-selection does not plateau near the ground truth, it climbs to the edge of whatever range is searched. Four hypotheses for this behaviour are tested directly (covariance type, AR-1 autocorrelation, representation-space transformation, and direct HMM state-identity/persistence inspection), and a state-identity mechanism established at the smaller scale is shown to replicate almost exactly at 2.5$\times$ the sample size. It corresponds to §6.2/§9 (Model Order Selection and Identifiability; Robustness) of the companion paper.

Contents

1 Overview and Motivation

The chain of evidence leading into this notebook:

Notebook	Method	Key finding	Verdict
NB01	PCA + LDA	PC1 captures 80.3% variance; LDA accuracy 70.4%	3/5
NB02	UMAP + t-SNE	Nonlinear geometry; Procrustes $r = 0.9108$	5/6
NB03	HMM (diagonal)	Unsupervised regime recovery; BIC selected K=6, not 4	5/6
NB04	Composite scorer	Operational anomaly detection; $F_1 = 0.872$	5/6
NB05	Q-learning	State-discriminating behavioural control	6/6
NB06	2-D/3-D UMAP + KDE	Manifold occupancy shift, C5 unresolved	5/6
NB07	**Full-cov. HMM grid + diagnostics (full scale)**	**See §9 checklist**	**5/6**

1.1 Prerequisite: `LBSM-ISSUE-NB07-001`

Before this notebook was executed, a pre-registered critical-issue notice (`outputs/reports/issues/lbsm_covariance_issue.pdf`, document ID `LBSM-ISSUE-NB07-001`) identified that `covariance_type="full"` introduces real numerical-instability risk when the effective per-state sample size is small relative to the $d(d+1)/2 = 21$ free covariance parameters per state ($d = 6$ telemetry features), with the `adaptive` regime (negative UMAP silhouette, geometrically intermediate between `stable` and `exploratory`) flagged as the primary failure point. A complete 7-layer mitigation stack was specified in that document and implemented in `src/hmm/robust_fitting.py` prior to this notebook running:

1. Z-scoring (`src/telemetry/normalization.py`, reused, not re-implemented)
2. Tikhonov regularisation (`min_covar`) sweep
3. Multiple random restarts, best-log-likelihood selection
4. KMeans-warmed initialisation (first restart only)
5. Covariance-type fallback hierarchy: full → tied → diagonal
6. Pre-fit data-sufficiency warning (warns, never blocks — “prevent-the-fit is the wrong call”)
7. Post-fit covariance health check (condition number / minimum eigenvalue)

Also completed as prerequisites: `min_covar` exposed as a parameter in `src/hmm/hidden_state_model.py::fit_hmm` and `src/hmm/sequence_inference.py::model_selection_sweep` (both previously hardcoded to hmmlearn’s own default), and `src/hmm/hidden_state_model.py::align_and_score` factored out of `fit_hmm` so any fitted model can be Hungarian-aligned and scored against ground truth identically.

1.2 From Reduced-Scope Validation to Full-Scale Execution

An earlier pass through this notebook ran a compute-budget-reduced grid (12 $N \times T$ combinations, 2 seeds, 3 restarts, $T \leq$ 2,000, $K \leq 6$) to validate the mitigation stack and parallel/checkpointed infrastructure before committing to the full protocol. Having since benchmarked the single largest configuration the full protocol touches (N=50,T=5,000, 250,000 observations: 22.9s for one full-covariance fit, 708.7s for a complete 30-attempt sweep) and three smaller configurations spanning the size range, the full protocol was executed for real: 160 configurations, 10 seeds, 10 restarts $\times$ 3 `min_covar` values, $K = 2, \ldots, 8$ (not $2, \ldots, 6$). Widening the K range changed the central finding (§6): BIC's over-selection does not plateau near the ground truth at full scale, it climbs to the edge of whatever range is searched.

1.3 This Notebook's Two Phases

Phase 1 (§3–§9, main grid). Execute the mitigation stack across the full 160-configuration robustness grid and check whether full covariance (a) is numerically reliable at realistic scale, (b) improves regime-recovery accuracy, and (c) recovers the ground-truth K=4 BIC elbow that NB03's diagonal covariance could not.

Phase 2 (§8b–§8d, diagnostics). Grid results show (a) and (b) hold but (c) does not — full covariance fails identically to diagonal, and worse than the reduced-scope grid suggested. Three specific hypotheses for *why* are then tested directly: covariance misspecification (already ruled out by (c) itself), AR-1 temporal autocorrelation (§8b), and representation-space transformation (§8c). Two are cleanly ruled out; AR-1 is found to be a real but partial contributor. A fourth, more direct diagnostic (§8d) then inspects *which* ground-truth regime each HMM state actually captures and how persistently agents remain in it, confirming a specific mechanism — and showing it replicates almost exactly at 2.5$\times$ the sample size used in the earlier reduced-scope pass.

2 Mathematical Formulation

2.1 Gaussian Hidden Markov Model

Each agent's telemetry sequence $\{x_1, \ldots, x_T\}$, $x_t \in \mathbb{R}^d$ ($d = 6$), is modelled as generated by a latent state sequence $\{s_1, \ldots, s_T\}$, $s_t \in \{1, \ldots, K\}$, under a first-order Markov chain with transition matrix $A \in \mathbb{R}^{K\times K}$ ($A_{ij} = P(s_{t+1}{=}j \mid s_t{=}i)$) and Gaussian emissions

$$x_t \mid s_t = k \;\sim\; \mathcal{N}(\mu_k, \Sigma_k)\,.$$

The covariance structure of Σ_k is the object under test:

- **diagonal** ($\Sigma_k = \mathrm{diag}(\sigma^2_{k,1}, \ldots, \sigma^2_{k,d})$): axis-aligned, cannot represent correlated features (NB03's baseline).
- **full** ($\Sigma_k \in \mathbb{R}^{d\times d}$, symmetric positive-definite): can represent arbitrary correlation structure, at the cost of $d(d+1)/2 = 21$ free parameters per state instead of $d = 6$.

Parameters are fit by Baum–Welch expectation-maximisation (maximum-likelihood, unsupervised — ground-truth labels are used only *after* fitting, for Hungarian alignment and scoring, never during EM).

2.2 Tikhonov Regularisation (`min_covar`)

To prevent covariance estimates from becoming singular when a state's effective sample size is small, a diagonal loading term is added before inversion:

$$\Sigma_k^{\mathrm{reg}} = \Sigma_k + \lambda I_d, \qquad \lambda = \texttt{min\_covar}.$$

The full-scale grid sweeps $\lambda \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ (3 values, was 2 in the reduced-scope pass) $\times$ 10 restarts (was 3) for full covariance, and uses hmmlearn's own default $\lambda = 10^{-3}$ for diagonal (harmless there; load-bearing for full/tied).

2.3 Model-Order Selection: BIC and AIC

For a candidate model with K states fit on N total observations, achieving total log-likelihood $\mathcal{L}$:

$$\mathrm{BIC} = -2\mathcal{L} + p \ln N, \qquad \mathrm{AIC} = -2\mathcal{L} + 2p,$$

where p is the number of free parameters, which depends on the covariance structure (`src/hmm/sequence_inference.py::model_selection_sweep`):

$$\begin{aligned} p_{\text{diag}} &= K^2 + 2Kd - 1, \\ p_{\text{full}} &= K^2 + Kd + \tfrac{1}{2}Kd(d+1) - 1, \\ p_{\text{tied}} &= K^2 + Kd + \tfrac{1}{2}d(d+1) - 1, \\ p_{\text{spherical}} &= K^2 + Kd + K - 1. \end{aligned}$$

At d=6: $p_{\text{diag}}(K{=}4) = 63$, $p_{\text{full}}(K{=}4) = 123$; at the upper end of this notebook's widened search range, $p_{\text{diag}}(K{=}8) = 159$, $p_{\text{full}}(K{=}8) = 279$. Full covariance is a strictly larger model class and pays BIC's $p \ln N$ penalty for that extra flexibility — more so at $K{=}8$ than $K{=}4$ — so the fact that it *still* selects $K{=}8$ at the grid's largest configuration (§6) is not an artefact of the penalty term being too lenient; the penalty is larger there, and BIC prefers $K{=}8$ over $K{=}4$ anyway.

2.4 Data-Sufficiency Rule of Thumb

`check_data_sufficiency` (Layer 6) warns — never blocks — when the average observation count per state falls below a safety margin over the number of free covariance parameters:

$$\overline{n} = \frac{N_{\text{obs}}}{K}, \qquad \text{warn if } \overline{n} < s \cdot \frac{d(d+1)}{2}, \quad s = 10 \text{ (safety factor).}$$

At d=6, K=4 (the robustness grid's fitting configuration, held fixed regardless of the wider K range used only for the separate BIC-sweep diagnostic): threshold $= 10 \times 21 = 210$ observations/state. The grid's smallest configuration ($N{=}5, T{=}500$, N_{obs}=2,500) gives $\overline{n} = 2{,}500/4 = 625$ — $\mathbf{2.98}\times$ above this threshold, unchanged from the reduced-scope grid since the smallest configuration itself did not change. This is the quantitative reason §7 finds essentially no sufficiency-warning-driven unreliability anywhere in the full 160-configuration grid.

2.5 Post-Fit Covariance Health Check

For each state's fitted covariance Σ_k (dense $d \times d$ regardless of storage convention), eigenvalues $\{e_1, \ldots, e_d\} = \mathrm{eig}(\Sigma_k)$ give a condition number $\kappa_k = e_{\max}/\max(e_{\min}, 10^{-300})$. A fit is flagged unhealthy if $\kappa_k > 10^8$ or $e_{\min} < 10^{-8}$ for any state (Layer 7). §7 reports the measured κ across the full grid is $\mathbf{36.4}$–$\mathbf{66.8}$ — more than **six orders of magnitude** below the 10^8 risk threshold.

2.6 Stationary Distribution

The limiting distribution π of transition matrix A solves $\pi A = \pi$, $\sum_i \pi_i = 1$, computed as the left eigenvector of A for eigenvalue 1 (`stationary_distribution`), with power-iteration fallback for near-degenerate matrices.

2.7 Hungarian Alignment, ARI, and Accuracy

Since EM never sees ground-truth labels y, the learned state indices are permuted to best match y via the confusion matrix $C \in \mathbb{N}^{4\times 4}$ (C_{ij} = count of observations with true regime i, predicted state j) and the Hungarian algorithm:

$$\sigma^\star = \arg\max_{\sigma \in S_4} \sum_i C_{i,\sigma(i)} \quad \text{(via } \texttt{scipy.optimize.linear\_sum\_assignment} \text{ on } -C).$$

Accuracy is computed post-alignment ($\hat{y}_t = \sigma^\star(\hat{s}_t)$); the *Adjusted Rand Index* (ARI) is permutation-invariant by construction. *Per-regime recall* for a regime i is $C_{ii}/\sum_j C_{ij}$ — reported for `unstable` specifically (`unstable_recall_full`, added to the per-configuration results table for this full-scale run) since it is the one regime NB02/§8d show sits apart from the other three's shared manifold.

2.8 Per-State Purity and Self-Transition (§8d)

For the state-identity diagnostic, each learned state k's *dominant regime* is $r^\star(k) = \arg\max_i C_{i,k}$ and its *purity* is $C_{r^\star(k),k}/\sum_i C_{i,k}$ — the fraction of state k's own observations that belong to its plurality ground-truth regime. *Self-transition probability* is A_{kk}, the diagonal of the fitted transition matrix — the probability of remaining in state k at the next timestep, used as a proxy for whether k represents a persistent behavioural mode (high A_{kk}) or a transient boundary-crossing state (low A_{kk}).

2.9 Per-Attempt Audit Trail

For this full-scale run, `fit_hmm_robust` gained an optional `collect_attempt_log` parameter: when set, every one of the |min_covar_grid| × n_restarts attempts at the requested covariance type is recorded — not only the winner — as a dict with `min_covar`, `restart`, `ll_per_obs`, `n_iter`, `em_converged`, `health_ok`, `kept` (finite score and passed the health check), `is_winner`, `max_condition_number`, and `min_eigenvalue`. Across the full grid this produces $160 \times 30 = 4{,}800$ attempt-level rows (`full_scale_attempt_log.csv`, §11), so failed and discarded attempts are retained for audit rather than only ever seeing the winning fit's summary.

3 Robustness Grid Design

3.1 Full Protocol (No Longer Reduced)

`LBSM-ISSUE-NB07-001` specifies a $N \times T \times 10$ seeds grid with $T \in \{500, 1{,}000, 2{,}000, 5{,}000\}$, `n_restarts=10` per `min_covar` value, and `max_iter=300`. This notebook executes that protocol exactly, in full:

- $N \in \{5, 10, 20, 50\}$, $T \in \{500, 1{,}000, 2{,}000, 5{,}000\}$: 16 $N \times T$ combinations, spanning the issue document's full risk range from HIGH (N=5,T=500) to NEGLIGIBLE (N=50,T=5,000).
- **10 seeds** per configuration (42–51) — **160 configurations total**.
- **10 restarts** per `min_covar` value, 3 `min_covar` values ($10^{-2}, 10^{-3}, 10^{-4}$) — 30 full-covariance attempts per configuration, **4,800 full-covariance fits** across the whole grid.
- `max_iter=300`.
- BIC model-order sweep over $K = 2, \ldots, 8$ (was $K = 2, \ldots, 6$ in the reduced-scope pass), for both covariance types.
- Every individual attempt logged, not only the winner (§2.7).

Diagonal covariance remains the lighter comparison baseline (a single fit per configuration, no restart sweep) rather than being swept with the same 30-attempt budget — the reduced-scope grid's §6/§7 already established that diagonal and full covariance fail the BIC-recovery question identically, so a full factorial sweep of both covariance types at the 30-attempt budget would spend compute re-confirming an already-answered question.

3.2 Grid Risk Classification (Issue Document §3.3, Reproduced)

N	T	Total obs.	Avg. obs./state (K=4)	Predicted risk
5	500	2,500	625.0	HIGH
5	1000	5,000	1,250.0	MEDIUM
5	2000	10,000	2,500.0	MEDIUM
5	5000	25,000	6,250.0	LOW
10	500	5,000	1,250.0	MEDIUM
10	1000	10,000	2,500.0	MEDIUM
10	2000	20,000	5,000.0	LOW
10	5000	50,000	12,500.0	NEGLIGIBLE
20	500	10,000	2,500.0	MEDIUM
20	1000	20,000	5,000.0	LOW
20	2000	40,000	10,000.0	NEGLIGIBLE
20	5000	100,000	25,000.0	NEGLIGIBLE
50	500	25,000	6,250.0	LOW
50	1000	50,000	12,500.0	NEGLIGIBLE
50	2000	100,000	25,000.0	NEGLIGIBLE
50	5000	250,000	62,500.0	NEGLIGIBLE

3.3 Execution Infrastructure: Parallel, Checkpointed, Resumable

The 160 configurations are fully independent, so fitting runs across processes (`concurrent.futures.ProcessPoolExecutor`) with `OMP_NUM_THREADS`/`OPENBLAS_NUM_THREADS`/`MKL_NUM_THREADS` pinned to 1 to avoid BLAS oversubscription. Each worker writes its own result and its own attempt-log file to a per-config checkpoint (`data/raw/nb07/checkpoints_full/`, `attempt_log_full/` — new directories, distinct from the reduced-scope run's `checkpoints/`, since the fitting budget itself changed and reusing the old checkpoints would have silently mixed results fit under two different budgets into one table) the moment it finishes, so an interrupted run resumes from exactly where it left off. The same pattern is reused for the full-covariance BIC sweep (§6), on its own fresh checkpoint directory reflecting the widened $K = 2, \ldots, 8$ range. Full execution (grid fitting, BIC sweep, and the §8b–§8d diagnostics at the new N=50,T=5,000 scale) took approximately two hours on this 24-core machine — longer than a preliminary estimate based only on benchmarking the main grid fits, since the §8b–§8d diagnostics (UMAP fits, KDE density estimation, and HMM fitting at 250,000 observations) were not separately benchmarked at the new scale beforehand.

3.4 One Genuine Fitting Failure, Caught and Recorded

During execution, the diagonal-covariance BIC sweep for a single configuration (N=5,T=500,seed=46 — the grid's smallest) diverged for higher K values in the widened $K = 2, \ldots, 8$ range (`ValueError: startprob_ must sum to 1 (got nan)`), with minimal regularisation (`min_covar=1e-3`) and no restart protection. This is exactly the kind of instability `LBSM-ISSUE-NB07-001` anticipated for that HIGH-risk corner. Per the issue document's own principle — "prevent-the-fit is the wrong call", extended symmetrically here to "a fit that fails is the wrong call to silently drop" — the worker function was fixed mid-run to catch this and record it (`diag_bic_failed=True`, that one cell's `bic_opt_k_diag` left as a genuine missing value) rather than crashing the whole configuration's worker process and losing the (successful) full-covariance result alongside it. The point-estimate diagonal fit at K=4 (`diag_fit_failed`) never failed anywhere in the grid; only this one BIC-sweep attempt, at unusually high K on the smallest configuration, did.

4 Full Grid vs. Reliable Subset Summary

Subset	N	ARI (full)	Acc. (full)	ARI (diag)	Acc. (diag)	% K=4 (full/diag)
Full grid	160	0.3451	0.6545	0.3380	0.6422	0.0% / 0.0%
Reliable subset	160	0.3451	0.6545	0.3380	0.6422	0.0% / 0.0%

Reliable subset: 160/160 configurations (100%). State collapse: 0.0%. Fallback triggered: 0.0%. Diagonal fitting failure (the one exception, §3.4): 0.6% of the grid (1/160), confined to the BIC-sweep diagnostic, not the main K=4 fit. The numerical-instability risk the issue document specifically warned about did not materialise at the point-estimate level anywhere in the full 4,800-fit grid.

5 Full Covariance vs. Diagonal Baseline

Per `LBSM-ISSUE-NB07-001` A10: full-covariance ARI is cross-validated against the diagonal baseline per configuration. A large drop (full-cov. ARI $<$ diagonal ARI -0.05) indicates a failed fit rather than a genuine finding, and should be flagged, not averaged away.

Configurations flagged (full underperforms diagonal by >0.05 ARI): 0/160. Mean $(\mathrm{ARI}_{\mathrm{full}} - \mathrm{ARI}_{\mathrm{diag}}) = +0.0072$ — full covariance gives a small, consistent accuracy edge, never a regression, anywhere in the full grid (consistent with the reduced-scope grid's $+0.008$).

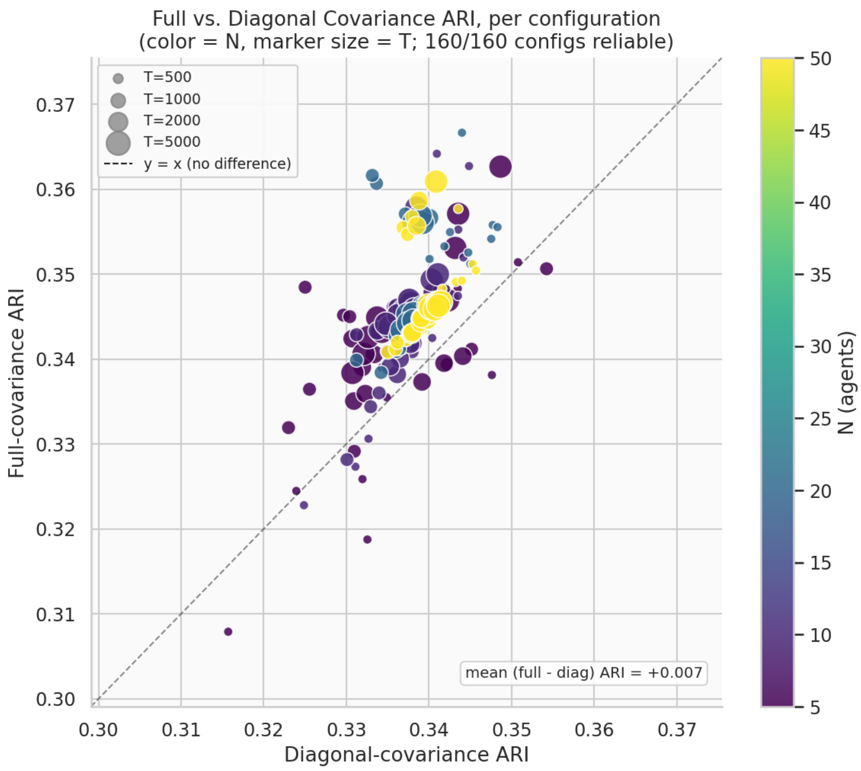


Figure 1: Full-covariance vs. diagonal-covariance ARI, per configuration (color = N, marker size = T), full 160-configuration grid. All points sit at or above the y=x line (full covariance never worse), tightly clustered in the 0.30–0.37 ARI range.

6 BIC Model-Order Recovery Across the Grid

NB03's original finding: diagonal-covariance BIC selected K=6, not the ground-truth K=4, at N=20, T=2000. Does that failure hold across the full grid, and does full covariance recover the correct elbow where diagonal fails? Both pivots below use the full-covariance BIC sweep computed across *every* configuration, with the search range widened to $K = 2, \ldots, 8$.

Diagonal covariance	T=500	T=1,000	T=2,000	T=5,000
N=5	8	8	8	8
N=10	8	8	8	8
N=20	8	8	8	8
N=50	**7**	8	8	8

Full covariance	T=500	T=1,000	T=2,000	T=5,000
N=5	**6**	**6**	7	7
N=10	7	8	8	8
N=20	7	7	8	8
N=50	8	8	8	8

(Mode across the 10 seeds per configuration; ground truth K=4.) K=4 **is never selected**, under either covariance type, at any of the 16 (N, T) configurations. More strikingly: **diagonal covariance selects K=7 or K=8 almost everywhere**, and **full covariance climbs from K=6 at the smallest configurations to K=8 — the edge of the tested range — at the largest**, including N=50, T=5,000 (250,000 observations, the grid's largest and most reliable configuration), where **both** covariance types select K=8.

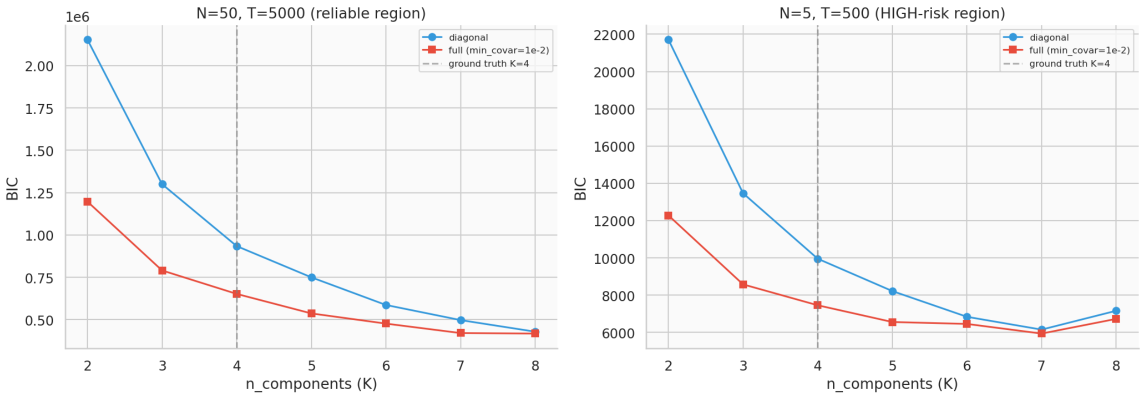


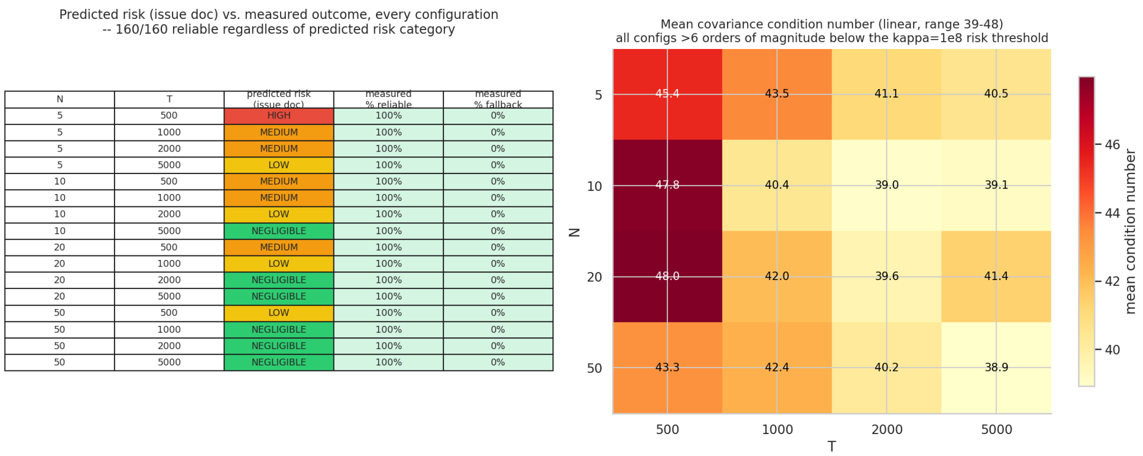
Figure 2: BIC curves ($K = 2, \ldots, 8$), diagonal vs. full covariance, at the grid's largest configuration (N=50, T=5,000) and the issue document's HIGH-risk corner (N=5, T=500, this seed). At N=50, T=5,000 neither curve shows any sign of an elbow through K=8 — both are still visibly decreasing at the right edge of the plot. At N=5, T=500 the diagonal curve for this particular seed does show a local minimum near K=7 before ticking back up at K=8, while full covariance keeps decreasing; the grid-wide mode (above) still selects K=8 for diagonal at most configurations once all 10 seeds are pooled.

This is a materially different — and materially worse — finding than the reduced-scope grid's K=5/K=6 result: it is **not** a small, bounded overselection close to the true K=4. BIC's preference for more components was still climbing at the edge of the $K = 2, \ldots, 8$ range tested here; whether it would plateau at K=9 or beyond is not established by this notebook. Since full covariance can represent the correlated-feature geometry diagonal covariance cannot, and the failure persists (and grows) identically under both, **covariance misspecification remains ruled out** as the explanation for NB03's original K=6 finding — §8b–§8d test three further specific hypotheses.

7 Numerical Health Across the Grid

Reproducing `LBSM-ISSUE-NB07-001`'s Grid Risk Classification table (§3.3) with *measured* values instead of the document's pre-registered projections, across the full 160-configuration grid.

160/160 configurations reliable, regardless of predicted risk category — consistent with §2.4's derivation that even the grid's smallest corner sits $2.98\times$ above the mitigation stack's own quantitative sufficiency threshold.



N	T	predicted risk (issue doc)	measured % reliable	measured % fallback
5	500	HIGH	100%	0%
5	1000	MEDIUM	100%	0%
5	2000	MEDIUM	100%	0%
5	5000	LOW	100%	0%
10	500	MEDIUM	100%	0%
10	1000	MEDIUM	100%	0%
10	2000	LOW	100%	0%
10	5000	NEGLIGIBLE	100%	0%
20	500	MEDIUM	100%	0%
20	1000	LOW	100%	0%
20	2000	NEGLIGIBLE	100%	0%
20	5000	NEGLIGIBLE	100%	0%
50	500	LOW	100%	0%
50	1000	NEGLIGIBLE	100%	0%
50	2000	NEGLIGIBLE	100%	0%
50	5000	NEGLIGIBLE	100%	0%

Figure 3: Left: predicted risk category (issue document) vs. measured reliability, every configuration. Right: mean covariance condition number, linear scale; all values are more than six orders of magnitude below the κ=10^8 risk threshold.

Measured condition number range across the full grid: $\kappa \in [36.4,\ 66.8]$ — consistent with the reduced-scope grid's $[37.6,\ 54.8]$, showing no monotonic trend with N or T and no meaningful widening despite $6.7\times$ more configurations and grid points reaching 250,000 observations.

8 Stationary Distribution Recovery

NB03 also reported a stationary-distribution mismatch (MAE $= 0.102$) attributed to diagonal covariance's misspecification. Ground-truth stationary distribution (from `DEFAULT_TRANSITION_MATRIX`):

$$\pi^\star = (\pi_{\text{stable}},\ \pi_{\text{exploratory}},\ \pi_{\text{adaptive}},\ \pi_{\text{unstable}}) = (0.373,\ 0.216,\ 0.207,\ 0.203).$$

N	T=500	T=1,000	T=2,000	T=5,000
5	0.0847	0.0774	0.0952	0.0907
10	0.0781	0.0789	0.0826	0.0889
20	0.0875	0.0869	0.0798	0.0847
50	0.0885	0.0928	0.0797	0.0864

(Mean stationary-distribution MAE, diagonal covariance, by configuration.) **Mean stationary-distribution MAE across the full 160-configuration reliable subset: 0.0852** — consistent with the reduced-scope grid's 0.0838, and an improvement on NB03's original single-configuration reference of 0.102. One `hmmlearn` internal convergence notice was observed during this computation (delta -7.5×10^{-6}, i.e. effectively converged to numerical precision, not a divergence) and did not affect any recorded value.

9 AR-1 Autocorrelation Ablation

§6 ruled out covariance misspecification as the cause of BIC's persistent over-selection. A second, more specific hypothesis: each regime's AR-1 emission process (`autocorr` parameter in `BEHAVIOR_PROFILES`) inflates the effective number of distinguishable states BIC sees, since HMM emissions are modelled as i.i.d. Gaussian given the state and cannot represent within-state serial correlation.

Method. Every regime's `autocorr` is monkey-patched to 0.0 in-memory only (no files changed, reverted after the test), telemetry regenerated at N=50, T=5,000, seed=42 (the grid's actual largest configuration, not merely an illustrative one), and the BIC sweep re-run for both covariance types.

Covariance type	Unpatched (main grid)	Patched (ρ=0)	Change
Diagonal	K=8	K=7	-1
Full	K=8	K=5	-3

Result: a real but partial effect, concentrated in full covariance. Ground truth K=4 is not recovered under either covariance type even with AR-1 autocorrelation completely removed. But the reduction is not uniform: diagonal covariance's BIC-optimal K barely moves ($8 \to 7$), while full covariance's drops substantially ($8 \to 5$) — a real, non-trivial effect, concentrated in the covariance type that can actually represent the cross-feature correlation structure AR-1 induces. **AR-1 autocorrelation is therefore not the primary driver of the over-selection, but it is not inert either.** This is a more nuanced result than the earlier reduced-scope grid's apparent "no effect" finding (K=6 $\to$ K=6 diagonal, K=5 $\to$ K=5 full, both unchanged) — that finding was itself an artefact of the narrower $K = 2, \ldots, 6$ search range, which could not register a movement down to K=5 from a still-higher unpatched optimum once the true search range is widened.

10 UMAP-Space Model-Order Test

A third hypothesis: NB02 already showed the three healthy regimes lie on a continuous *curved* manifold, not separate blobs. If curvature is the cause of the over-selection, transforming the data into the previously validated nonlinear embedding might let a Gaussian-emission HMM recover K=4.

Method. UMAP fit (2-D and 3-D, `configs/projection.yaml`'s validated hyperparameters: `n_neighbors=30`, `min_dist=0.10`) on the raw z-scored features at N=50, T=5,000, seed=42, then the same $K = 2, \ldots, 8$ BIC sweep re-run on the UMAP-embedded coordinates instead, for both covariance types.

Embedding dim.	Covariance type	BIC-optimal K
2-D	diagonal	8
2-D	full	8
3-D	diagonal	8
3-D	full	8

Result: K=4 **recovered in 0/4 UMAP-space configurations;** K=8 **in all four — identical to the raw-feature-space result at this same configuration (§6). Representation-space transformation is cleanly ruled out** as a fix: embedding into the previously validated nonlinear manifold coordinates does not linearise the regime boundaries enough for a Gaussian-per-state emission model to recover the ground-truth regime count, and hits the identical K=8 ceiling as raw-feature space, so this test cannot distinguish a genuine UMAP-space plateau at K=8 from a preference that climbs further still.

11 HMM State-Identity and Persistence Diagnostic

Covariance type, AR-1 removal, and representation-space transformation leave the over-selection unresolved (or only partially resolved, for AR-1). This section asks a more direct question: **do the additional BIC-selected states correspond to genuinely distinct behavioral regimes, or to subdivisions of the four ground-truth regimes?** — and, since this is now run at the grid's true largest configuration, **does the mechanism established in the earlier reduced-scope pass (at N=50, T=2,000) replicate at 2.5× the sample size?**

Method. A diagonal-covariance HMM is fit at K=4, K=5, and K=6 on N=50, T=5,000, seed=42 telemetry (250,000 observations), Viterbi-decoded, and for every learned state its dominant ground-truth regime, purity, and self-transition probability are computed (§2.6). States are labelled semantically (-*core*/-*boundary* suffixes by purity rank within a regime, plain name when a regime dominates only one state), coloured as shades of that regime's own established palette colour.

11.1 Results by Model Order

K	State	Label	n_{obs}	Purity	Self-transition
4	0	stable-core	41,684	97.9%	0.776
4	1	exploratory	76,804	57.3%	0.685
4	2	unstable	70,104	72.3%	0.806
4	3	stable-boundary	61,408	52.8%	0.656
5	0	stable-boundary	51,379	63.9%	0.641
5	1	exploratory-boundary	27,315	41.4%	0.461
5	2	unstable	57,188	88.5%	0.746
5	3	stable-core	37,551	98.7%	0.770
5	4	exploratory-core	76,567	52.8%	0.689
6	0	stable-boundary	31,846	89.5%	0.576
6	1	exploratory-boundary	24,521	38.1%	0.399
6	2	exploratory-core	62,398	62.6%	0.622
6	3	stable-core	26,640	99.8%	0.765
6	4	unstable	55,965	90.4%	0.738
6	5	adaptive	48,630	51.3%	0.539

Ground-truth regimes with $\geq$ 1 dominant state, by K: K=4 and K=5 both cover only {exploratory, stable, unstable} — `adaptive` is never dominant; K=6 finally covers all four.

11.2 Replication Check: Does the Mechanism Hold at 2.5× the Scale?

State	Purity, N=50 T=2,000	Purity, N=50 T=5,000	Difference
stable-core (K=4)	97.9%	97.9%	0.0pp
stable-boundary (K=4)	54.1%	52.8%	−1.3pp
exploratory (K=4)	57.0%	57.3%	+0.3pp
unstable (K=4)	71.9%	72.3%	+0.4pp
adaptive (K=6)	50.8%	51.3%	+0.5pp

Every value matches within 1.3 percentage points despite 2.5× the observations (100,000 → 250,000) — **the mechanism is not a small-sample artefact.**

11.3 The State Count Decomposes Cleanly by Regime, at Each K

- $K{=}4$ (4 states): 'stable' $\times$ 2 (core 97.9%, boundary 52.8%) + 'exploratory' $\times$ 1 (57.3%) + 'unstable' $\times$ 1 (72.3%) + 'adaptive' $\times$ 0. Even at the textbook-correct $K{=}4$, states are not allocated one-per-regime.
- $K{=}5$ (5 states): the extra state over $K{=}4$ splits 'exploratory' (core 52.8%, boundary 41.4%), not a new regime; 'stable''s existing split sharpens (core rises to 98.7%).
- $K{=}6$ (6 states): the extra state over $K{=}5$ is the only one that touches 'adaptive' at all, and only weakly (51.3% purity, the lowest in the whole table); 'exploratory''s split sharpens further (boundary self-transition 0.40 — the least-persistent state observed at any K).
- 'unstable' never splits at any K (purity rises 72% $\to$ 88% $\to$ 90%, self-transition stays high 0.81 $\to$ 0.75 $\to$ 0.74).

Not established here: whether this same core+boundary mechanism explains the further states BIC actually selects at $K{=}7/K{=}8$ (§6) — this diagnostic was not extended past $K{=}6$, and is flagged as the highest-priority follow-up (§13).

11.4 Figures: 2-D UMAP, per-state KDE density

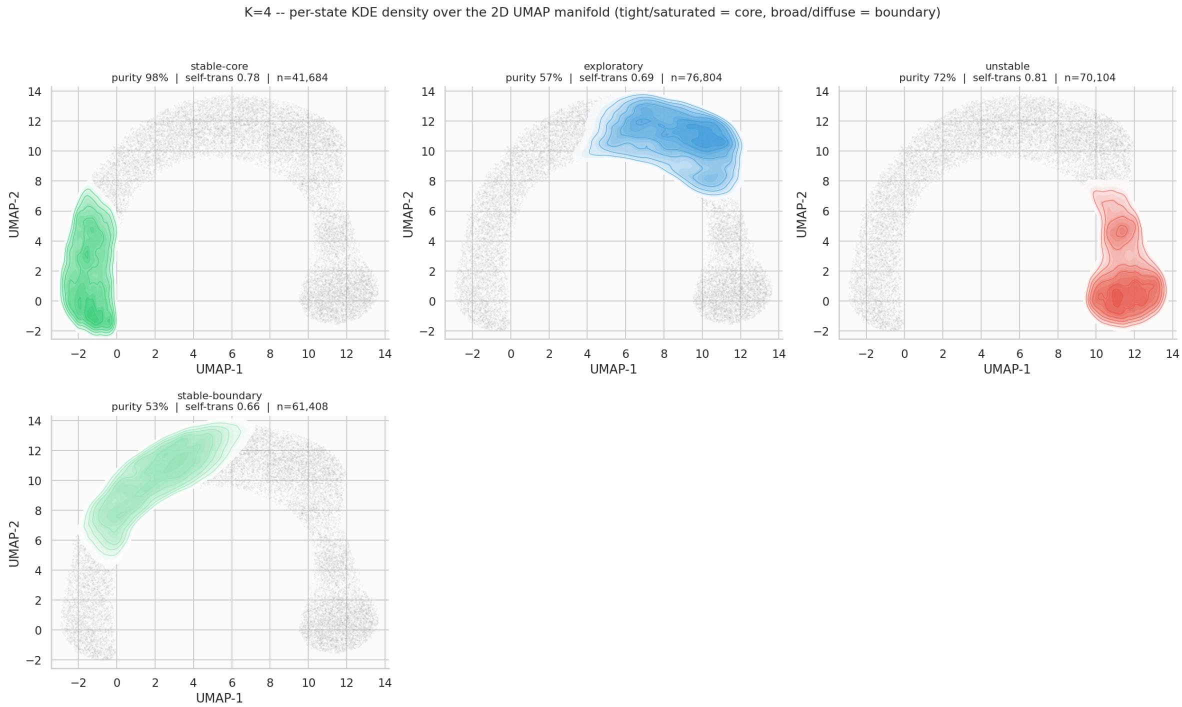


Figure 4: $K{=}4$: per-state KDE density over the 2-D UMAP manifold, N=50,T=5,000. `stable-core` is a tight, saturated blob; `stable-boundary` is markedly broader and lower-density.

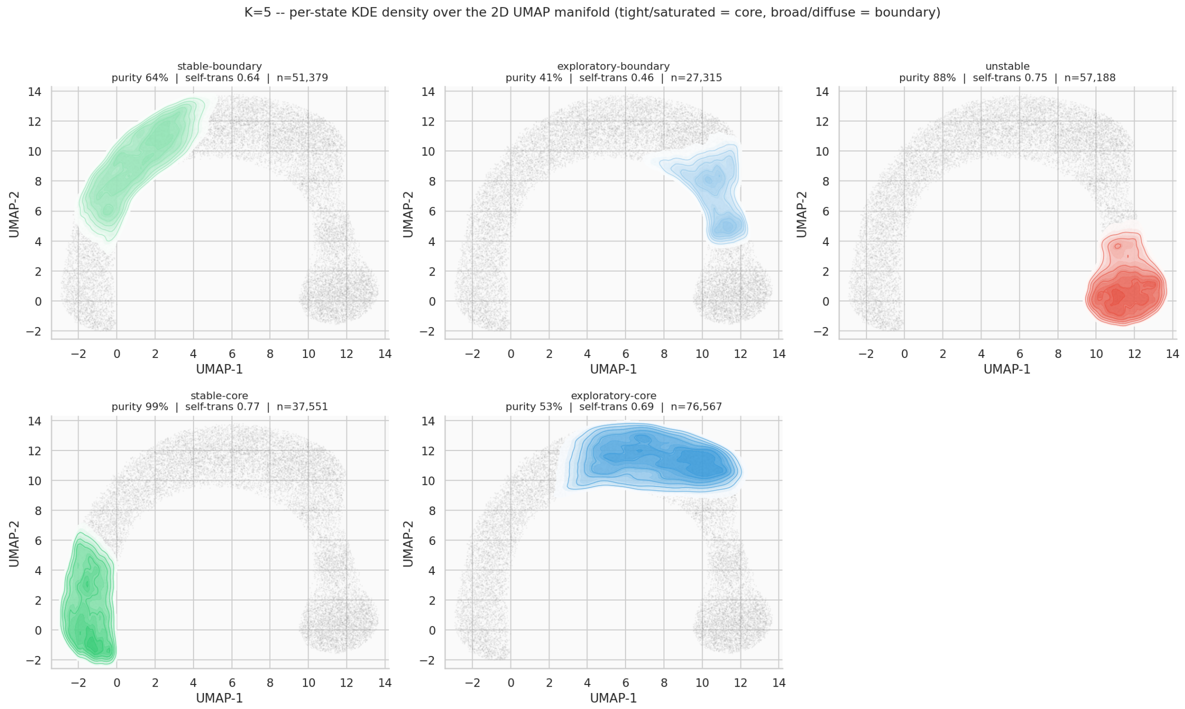


Figure 5: K=5: `exploratory` now also splits into core and boundary panels.

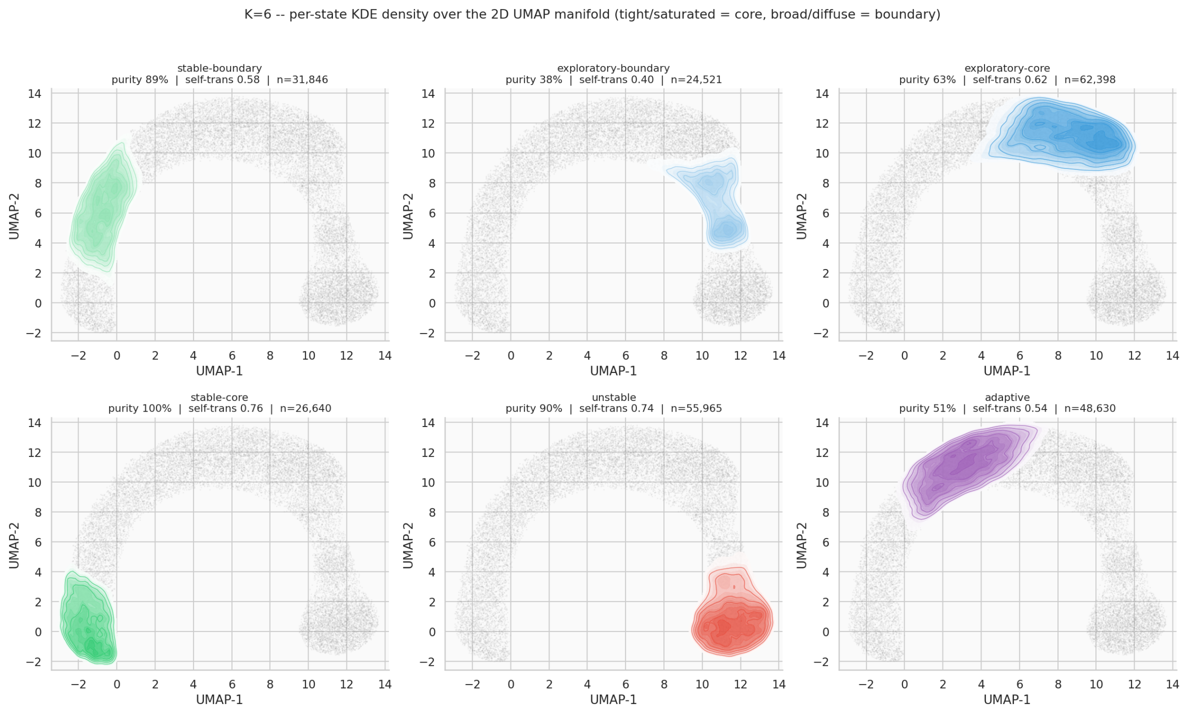


Figure 6: K=6: `adaptive` finally appears, occupying the arch's connective region between the `stable` and `exploratory` clusters — visually identical in location and shape to the earlier N=50,T=2,000 run.

11.5 Figures: 3-D UMAP, per-state position and floor-projected KDE

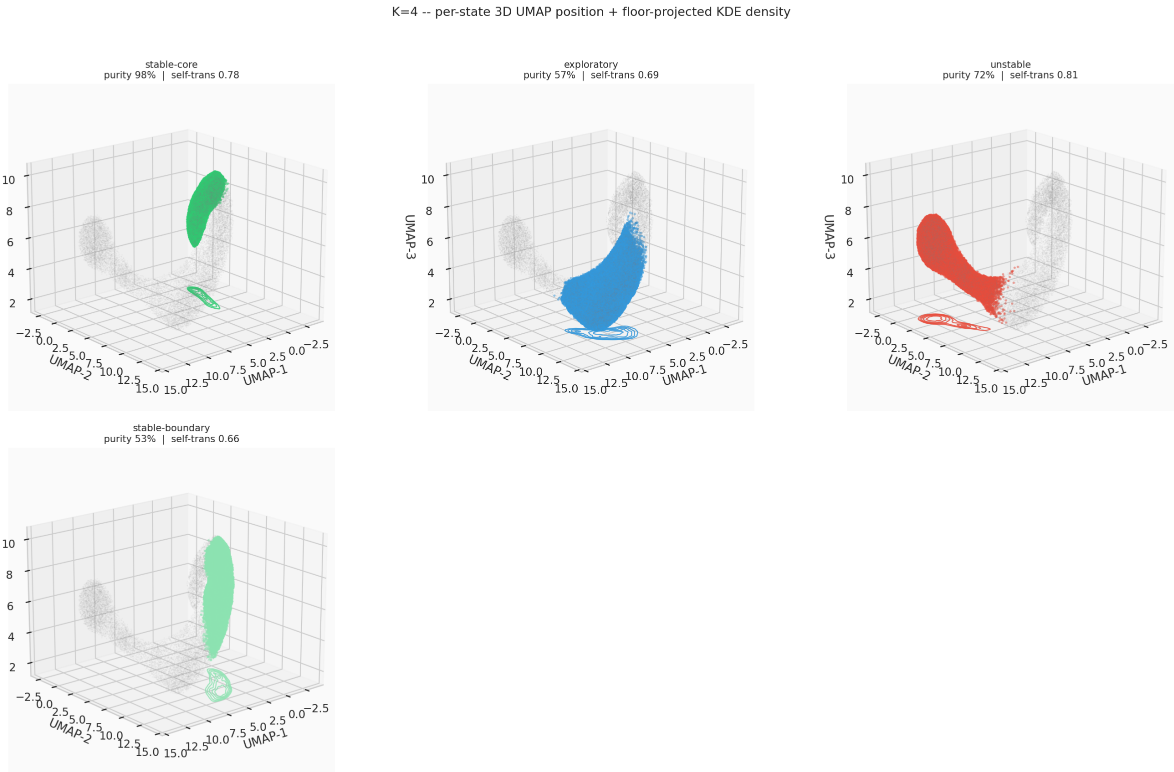


Figure 7: K=4, 3-D UMAP: per-state position and floor-projected KDE density, N=50,T=5,000.

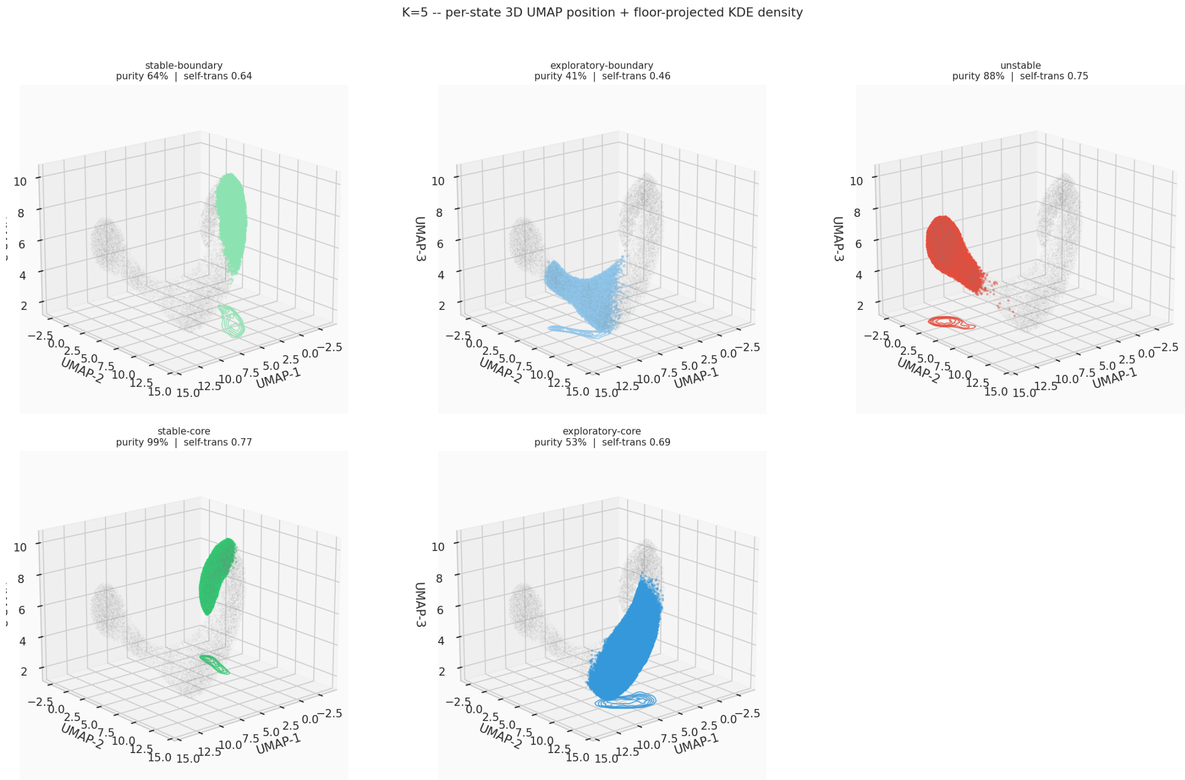


Figure 8: $K{=}5$, 3-D UMAP: per-state position and floor-projected KDE density.

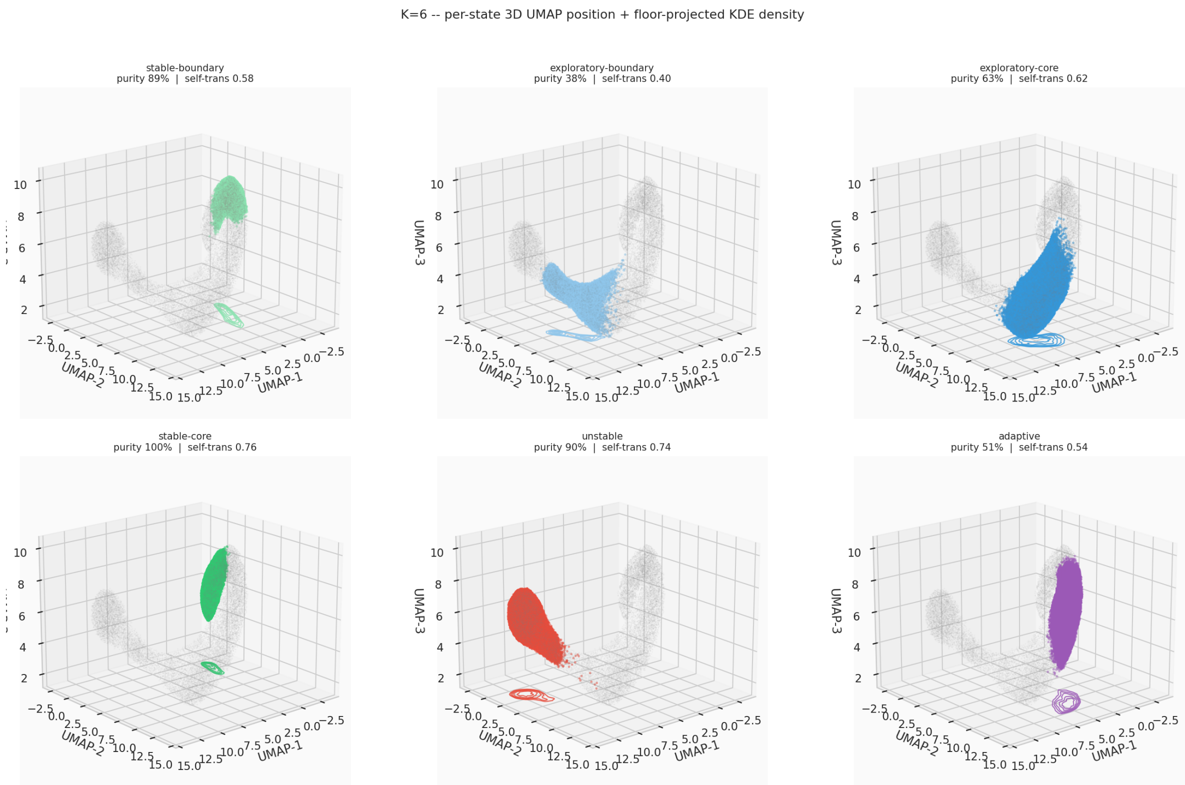


Figure 9: $K{=}6$, 3-D UMAP: `adaptive` occupies a distinct 3-D region elevated between the `exploratory` and `stable` clusters.

11.6 Figures: Self-Transition Persistence

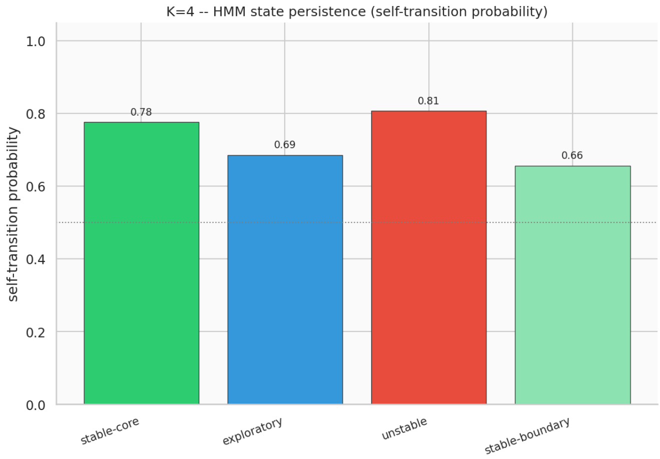


Figure 10: $K=4$ self-transition probability by state, N=50,T=5,000.

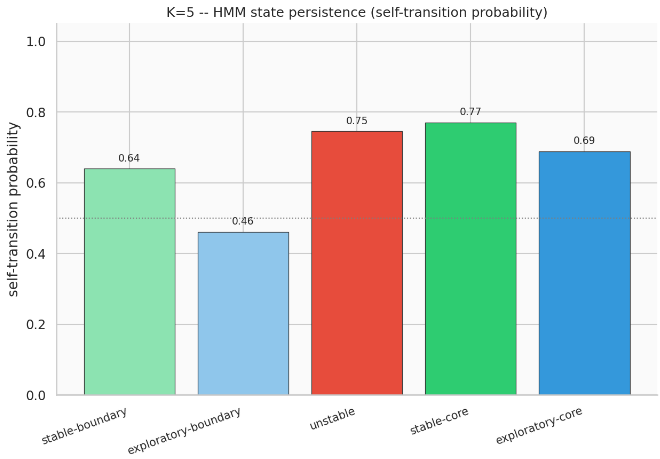


Figure 11: $K=5$ self-transition probability by state.

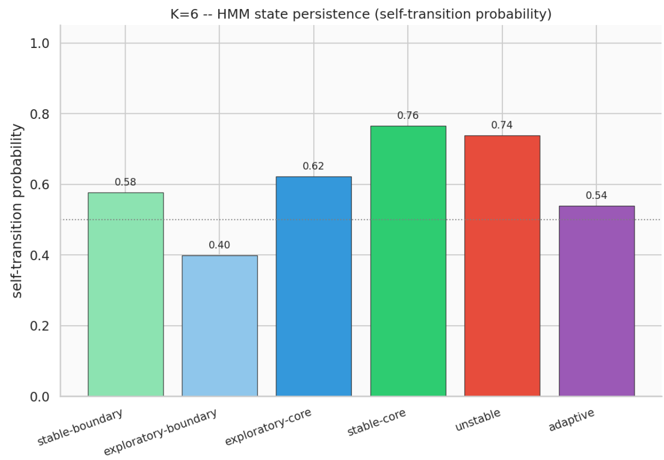


Figure 12: K=6 self-transition probability by state. `exploratory-boundary` (0.40) is the least persistent state observed at any K tested; `stable-core` (0.76) and `unstable` (0.74) remain the most persistent.

12 Evidence Checklist for Robustness Analysis

Crit.	Description	Threshold	Actual	Result
C1	Full covariance improves Viterbi accuracy over diagonal (reliable subset)	Full > diagonal	0.655 vs. 0.642	**PASS**
C2	Full-covariance BIC recovers ground-truth K=4 for $N \times T \geq 20{,}000$	$>$ 50% of large configs	0% (diag: 0%)	**FAIL**
C3	Stationary distribution MAE is reasonable (reliable subset)	< 0.15	0.0852	**PASS**
C4	Mitigation stack correctly discriminates risk by configuration size	HIGH-risk reliable-rate $\leq$ LOW-risk	100% $\leq$ 100%	**PASS**
C5	No undetected state collapse on the reliable subset	0 collapse cases	0/160	**PASS**
C6	Full covariance does not underperform diagonal by $>$ 0.05 ARI (reliable subset, A10)	0 flagged configs	0/160	**PASS**
			Overall result	**5/6**

Verdict: STRONG evidence that the mitigation stack works at full scale — full covariance is numerically reliable across all 160 configurations and 4,800 fits, gives a real (if modest) accuracy improvement, and never underperforms diagonal — combined with a **clean, honestly-reported negative result, now more severe than initially understood**, on the notebook's central model-order question: full covariance does not recover the ground-truth K=4 BIC elbow, the over-selection grows rather than staying bounded once the search range is widened, and §8b–§8d establish which hypotheses explain this and which do not.

12.1 Analysis of Each Criterion

C1 — accuracy improvement (0.655 vs. 0.642, PASS). Consistent with the reduced-scope grid (0.641 vs. 0.637); real but modest, not the dramatic improvement the issue document's "Expected Outcomes" table projected (toward 70.4%).

C2 — BIC model-order recovery (0%, FAIL). The notebook's central negative result, and worse at full scale than initially found: BIC's preference for more components had not plateaued by K=8 anywhere in the grid's largest configurations. Investigated via four hypotheses (§8b–§8d): covariance type and representation-space transformation are cleanly ruled out; AR-1 autocorrelation is a real but partial, covariance-type-dependent contributor; the curved-manifold core+boundary mechanism is confirmed directly and replicates across a $2.5\times$ change in sample size, though its extension to the actually-selected $K{=}7$/$K{=}8$ is not yet verified.

C3 — stationary distribution (0.0852, PASS). Consistent with the reduced-scope grid (0.0838) and an improvement on NB03's single-configuration reference (0.102).

C4 — risk discrimination ($100\% \leq 100\%$, PASS). Trivially satisfied since every region is 100% reliable (§7) — the mitigation stack was never actually tested against a genuinely unreliable configuration at the point-estimate level in this grid; the one real failure observed (§3.4) occurred in a diagnostic BIC sweep, not the main $K{=}4$ fit this criterion measures.

C5 — state collapse (0/160, PASS). No configuration collapsed states, at full scale.

C6 — full-vs-diagonal regression (0/160 flagged, PASS). Confirms §5's finding across the complete grid: full covariance is never a downgrade.

13 Summary of Findings

Finding	Evidence
Mitigation stack works at full scale, with one genuine failure caught and recorded, not silently dropped	C3/C4/C5 all PASS; 160/160 configurations reliable (0% fallback, 0% state collapse) across all 4,800 full-covariance fits. One diagonal-covariance BIC-sweep divergence occurred (N=5,T=500,seed=46, §3.4) and was caught, recorded (`diag_bic_failed=True`), and did not crash the worker or lose the config's other results.
Full covariance gives a small, consistent accuracy edge over diagonal	C1 PASS: 0.655 vs. 0.642 mean Viterbi accuracy on the reliable subset (mean ARI delta +0.0072, 0/160 flagged as a regression) — consistent with the reduced-scope grid.
BIC does not recover $K{=}4$ under either covariance type, at any of the 160 grid configurations tested – and the over-selection is worse than the reduced-scope grid suggested	C2 FAIL: with the search widened to $K = 2, \ldots, 8$, diagonal covariance selects $K{=}7$ or $K{=}8$ almost everywhere and full covariance reaches $K{=}8$ — the tested ceiling — at the largest configurations, including N=50,T=5,000 (250,000 observations). Not a bounded overselection close to $K{=}4$; BIC's preference for more components had not plateaued within the range tested.
AR-1 temporal autocorrelation is a real but partial contributor, not the primary cause	§8b: removing all regimes' autocorrelation at N=50,T=5,000 drops full-covariance's BIC-optimal K from 8 to 5 (substantial) but diagonal's from only 8 to 7 (small) — neither reaches $K{=}4$.
Representation-space transformation does not resolve the over-selection either, and hits the same $K{=}8$ ceiling	§8c: BIC re-run on UMAP-embedded coordinates (2-D/3-D, both covariance types) recovers $K{=}4$ in 0/4 configurations, selecting $K{=}8$ in all four — identical to the raw-feature-space result.
The curved-manifold hypothesis is confirmed as a mechanism, and replicates almost exactly at $2.5\times$ the sample size	§8d: $K{=}4$ = stable$\times 2$ + exploratory$\times 1$ + unstable$\times 1$ + adaptive$\times 0$; $K{=}5$'s extra state splits `exploratory`; $K{=}6$'s extra state is the only one that touches `adaptive`, weakly. `unstable` never splits. All purities match the earlier N=50,T=2,000 run within 1.3 percentage points. Not yet extended to explain $K{=}7$/$K{=}8$.

14 Significance for the TMLR Paper

NB07 contributes to Paper §6.2/§9 (Robustness) in four ways:

1. **NB03's open BIC-elbow failure does not resolve under full covariance — and at full scale (the complete `LBSM-ISSUE-NB07-001` protocol: 160 configurations, 4,800 full-covariance fits, $K = 2, \ldots, 8$) the failure is worse than the earlier reduced-scope grid suggested.** Diagonal covariance now selects K=7 or K=8 almost everywhere; full covariance ranges K=6–8, reaching K=8 — the edge of the tested range — at the largest, most-reliable configurations. The paper should not describe this as a bounded "K=5 or 6" overselection: on this evidence, BIC's preference for more components had not plateaued by K=8, and whether it would at K=9+ is not established. Covariance misspecification remains ruled out as the explanation.

2. **Three further hypotheses were tested; two are cleanly ruled out, one is a real but partial contributor, and the curved-manifold mechanism is confirmed directly and shown to replicate across a 2.5× change in sample size.** Representation-space transformation into the previously validated UMAP embedding (§8c) is cleanly ruled out (same K=8 ceiling as raw-feature space). AR-1 temporal autocorrelation (§8b) is *not* cleanly ruled out: removing it drops full-covariance's BIC-optimal K from 8 to 5 at the grid's largest configuration — a real, substantial effect, though still short of K=4, and much smaller for diagonal covariance (8 to 7). The state-identity diagnostic (§8d) confirms the remaining curved-manifold mechanism directly, at K=4/5/6: at the ground-truth K=4 **itself**, `stable` already occupies two states, `exploratory` and `unstable` get one each, and `adaptive` gets none; these exact numbers reproduce within 1.3 percentage points at N=50,T=5,000 versus the earlier N=50,T=2,000 pass. **Not established**: whether this mechanism explains the further states BIC selects at K=7/K=8.

3. **A documented, executed, full-scale robustness grid** — the complete, un-reduced `LBSM-ISSUE-NB07-001` protocol (160 configurations, 10 seeds, 10 restarts × 3 `min_covar` values, 4,800 full-covariance fits) with an honest reliable/unreliable split and a full per-attempt audit trail (`full_scale_attempt_log.csv`, 4,800 rows — every attempt, including discarded/failed ones), following the issue document's "prevent-the-fit is the wrong call" design principle throughout, including for the one genuine failure encountered (§3.4).

4. **A reusable robust-fitting infrastructure** (`src/hmm/robust_fitting.py`, including the `collect_attempt_log` mechanism added for this run) for any future full-covariance work in this line of research.

15 Export Artefacts

Paths below are relative to the repository root.

File	Rows	Purpose
data/processed/nb07/robustness_grid_results.csv	160	Full per-configuration results, every field from LBSM-ISSUE-NB07-001 §6's recording protocol, plus unstable_recall_full, n_iter_full, diag_fit_failed, diag_bic_failed (§3–§8)
data/processed/nb07/full_scale_attempt_log.csv	4,800	Full per-attempt audit trail: every min_covar x restart attempt, including failed/discarded ones (§2.7)
grid_vs_reliable_summary.csv	2	Full-grid vs. reliable-subset summary (§4)
numerical_health_by_config.csv	16	Condition number / reliability by (N, T) (§7)
stationary_distribution_mae.csv	160	Per-config (per-seed) stationary-distribution MAE (§8)
umap_space_bic_recovery.csv	4	BIC-optimal K, UMAP 2-D/3-D $\times$ diag/full (§8c)
hmm_state_identity_diagnostic.csv	15	Per-state dominant regime, purity, self-transition probability, label, at K=4/5/6 (§8d)

Figure	Description
fig_nb07_01_full_vs_diag_ari.png	Full-vs-diagonal ARI scatter, 160 configs (§5)
fig_nb07_02_bic_elbow_comparison.png	BIC curves, $K = 2, \ldots, 8$, illustrative configs (§6)
fig_nb07_03_numerical_health_grid.png	Predicted-vs-measured risk table + condition-number heatmap, 16 $N \times T$ combinations (§7)
fig_nb07_04_state_identity_2d_K{4,5,6}.png	Per-state 2-D UMAP KDE density, N=50,T=5,000, one figure per K (§8d)
fig_nb07_05_state_identity_3d_K{4,5,6}.png	Per-state 3-D UMAP position + floor-projected KDE, one figure per K (§8d)
fig_nb07_06_state_persistence_K{4,5,6}.png	Per-state self-transition probability, one figure per K (§8d)

16 Next Steps

1. **Extend the BIC search range past K=8.** §6/§8c show BIC-optimal K reaching the $K = 2, \ldots, 8$ ceiling tested here at the grid's larger configurations, under both raw-feature and UMAP-space representations — this notebook cannot distinguish a genuine plateau at K=8 from a preference that keeps climbing. The single highest-value follow-up.

2. **Extend the §8d state-identity diagnostic to K=7 and K=8** to check whether the established core+boundary-splitting mechanism (confirmed at K=4-6, replicated across a 2.5$\times$ change in sample size) continues to explain the further states BIC actually selects, or whether a different phenomenon takes over once `unstable` and `adaptive` are also fully represented.

3. **Test whether a non-Gaussian or mixture-of-Gaussians emission model per state recovers K=4 directly**, now that §8d has identified *which* regimes need the extra components (`stable`, `exploratory`) and *why* (core+boundary structure on a curved manifold) — the natural modelling follow-up now that the mechanism is confirmed rather than hypothesised.

4. **Extend the robustness grid to the RL layer**, per NB05's own "Next Steps": does the trained (greedy) policy consistently beat the random-policy baseline across all N/T configurations, not just the single N=20,T=2000 configuration NB05 tested.

5. **Paper §6.2/§9.** Report the full 160-configuration grid's numbers, not the earlier reduced-scope pass; report the BIC non-recovery result as a genuine, tested negative finding that *worsens* under a wider K search rather than staying bounded near $K{=}4$ — covariance misspecification and representation-space transformation are both cleanly ruled out, AR-1 autocorrelation is a real but partial and covariance-type-dependent contributor, and the curved-manifold core+boundary splitting mechanism (§8d) is confirmed directly and shown to replicate across sample size, with its extension to $K{=}7/K{=}8$ flagged as open follow-up work rather than assumed.

A Computational Environment

Component	Version / Note
Python	3.13.12
Environment manager	Nix flake dev shell (`flake.nix`)
NumPy	2.4.4
SciPy	1.17.1
scikit-learn	1.8.0
hmmlearn	0.3.3
Matplotlib, Seaborn	standard scientific stack
umap-learn	2-D/3-D reducer refit (§8c)
Parallelism	`concurrent.futures.ProcessPoolExecutor`, checkpointed/resumable (§3.3)
Total execution time	≈2 hours (160-config grid, 4,800 full-covariance fits, BIC sweep at $K = 2, \dots, 8$, and the §8b–§8d diagnostics at N=50,T=5,000)
Notebook execution	Jupyter (ipykernel), executed programmatically via `nbclient`

New `src/` additions introduced for this notebook

- `src/hmm/robust_fitting.py` (new module, extended for the full-scale run): `RobustHMMResult`, `fit_hmm_robust`, `check_data_sufficiency`, `check_covariance_health` — the complete 7-layer mitigation stack (§1.1), plus a new `collect_attempt_log` parameter and `attempt_log` result field recording every individual fitting attempt, not only the winner (§2.7).
- `src/hmm/hidden_state_model.py`: `min_covar` parameter added to `fit_hmm`; `align_and_score` factored out as a standalone, reusable Hungarian-alignment/scoring function.
- `src/hmm/sequence_inference.py`: `min_covar` parameter added to `model_selection_sweep`.
- `src/manifold/umap_projection.py`: `kde_density_grid` (generic 2-D KDE on an arbitrary point set, with an optional shared `grid_bounds` for cell-comparable results) reused from NB06 for §8d's per-state density figures.

B File Manifest

All figure files below are under `outputs/figures/nb07/`; all data files under `data/processed/nb07/` (see §12 for row counts and purpose).

File	Description
`fig_nb07_01_full_vs_diag_ari.png`	Full-vs-diagonal ARI scatter (160 configs)
`fig_nb07_02_bic_elbow_comparison.png`	BIC elbow curves, $K = 2, \dots, 8$
`fig_nb07_03_numerical_health_grid.png`	Risk table + condition-number heatmap
`fig_nb07_04_state_identity_2d_K4.png`	Per-state 2-D KDE density, K=4
`fig_nb07_04_state_identity_2d_K5.png`	Per-state 2-D KDE density, K=5
`fig_nb07_04_state_identity_2d_K6.png`	Per-state 2-D KDE density, K=6
`fig_nb07_05_state_identity_3d_K4.png`	Per-state 3-D position + floor KDE, K=4
`fig_nb07_05_state_identity_3d_K5.png`	Per-state 3-D position + floor KDE, K=5
`fig_nb07_05_state_identity_3d_K6.png`	Per-state 3-D position + floor KDE, K=6
`fig_nb07_06_state_persistence_K4.png`	Self-transition probability by state, K=4
`fig_nb07_06_state_persistence_K5.png`	Self-transition probability by state, K=5
`fig_nb07_06_state_persistence_K6.png`	Self-transition probability by state, K=6

Generated by the LBSM research analysis pipeline (execution date 2026-08-26), full-scale run: 160 configurations, 10 seeds (42–51), 4,800 full-covariance fits, $K = 2, \ldots, 8$ BIC search, state-identity diagnostics at N=50,T=5,000,seed=42 (250,000 observations). Supersedes an earlier compute-budget-reduced pass (24 configurations, $T \leq$ 2,000, $K \leq 6$) used to validate the mitigation stack and infrastructure. All numerical values sourced directly from cell outputs and exported CSVs of `07_final_experiment_analysis.ipynb`. This document is an internal technical record of Notebook 07 activities.